Interest Rate Swaps: Risk Model

2016

CME Group

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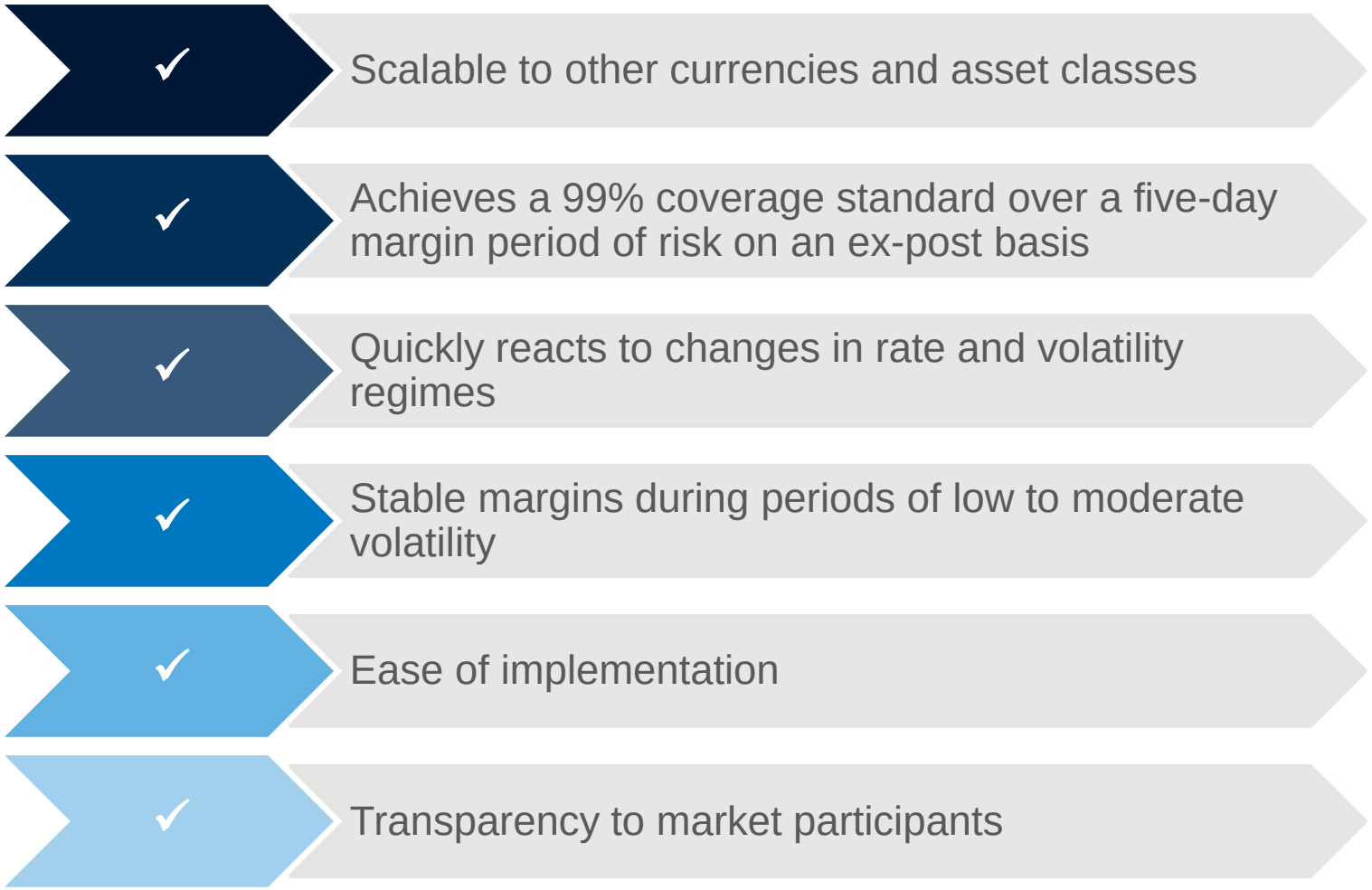
Products Available for Portfolio Margining

CME Clearing offers portfolio margining between the following exchange-traded interest rate derivatives, IRS, and swaptions:

- Eurodollar Futures
- U.S. Treasury Futures
  - 2-year
  - 5-year
  - 10-year
- U.S. Treasury Bond Future
- Ultra U.S. Treasury Bond Future
- Ultra 10-year U.S. Treasure Note Future
- 30-day Federal Funds Future
- U.S. MAC Swap Futures
  - 2-year
  - 5-year
  - 10-year
  - 30-year

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IRS Margin Model Goals



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Overview of IRS Margin Model

IRS Margin: HVaR and Add-ons

The CME Clearing IRS margin model consists of multiple components.

Market Risk Component: Historical Value-at-Risk ("HVaR")

- The IRS margin model generates scenarios for interest rate curves, futures prices, spot FX rates, and swaption implied volatilities of 5-day returns that are synchronized across multiple currencies, futures, and FX pairs
  - The model utilizes a rolling lookback period that begins on a given day in September 2008
- Historical returns are scaled based on an appropriate forecast of short term volatility: Exponentially Weighted Moving Average ("EWMA") volatilities
- Once the profit and loss distribution of the portfolio is determined based on the above scenarios, margin is sampled as 99.7 percentile of the loss distribution over a five-day margin period of risk
  - To determine the appropriate margin period of risk CME Clearing considers the liquidity and market characteristics for the given product
- Further, risk mitigation is employed by apportioning the USD margins down to each currency

Swaption Add-on: Additional add-ons are incorporated to capture swaption specific risks

Liquidity/Concentration Add-on: CME Clearing applies an add-on to portfolios that present significant liquidation risk within the close out period

- Liquidity costs are computed for each currency to reflect the expenses associated with liquidation
  - The liquidity add-on is calculated as a function of portfolio Greeks

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IRS Financial Safeguards

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IRS Financial Safeguards Overview

IRS Financial Safeguards

CME Group Exchange Rules define a default by a Clearing Member to be the failure of such Clearing Member to promptly discharge any obligation to the Clearing House or such Clearing Member becomes subject to any bankruptcy or insolvency proceedings

Guaranty Fund Sizing

- The IRS Guaranty Fund covers the potential loss caused by the simultaneous default of the two Clearing Members and their affiliates with the largest stress shortfall
- The IRS Guaranty Fund is re-sized on at least monthly basis or more frequently as market conditions warrant
  - The size of the IRS Guaranty Fund is calculated daily for risk management review
  - CME Clearing is permitted and able to re-size IRS Guaranty Fund intra-cycle, where necessary
- Assessment powers cover losses beyond the pre-funded resources comprising the IRS Guaranty Fund
  - Assessments for the IRS financial safeguards waterfall structure are sized to cover the stress loss arising from the default of the 3rd and 4th largest Clearing Members and their affiliates

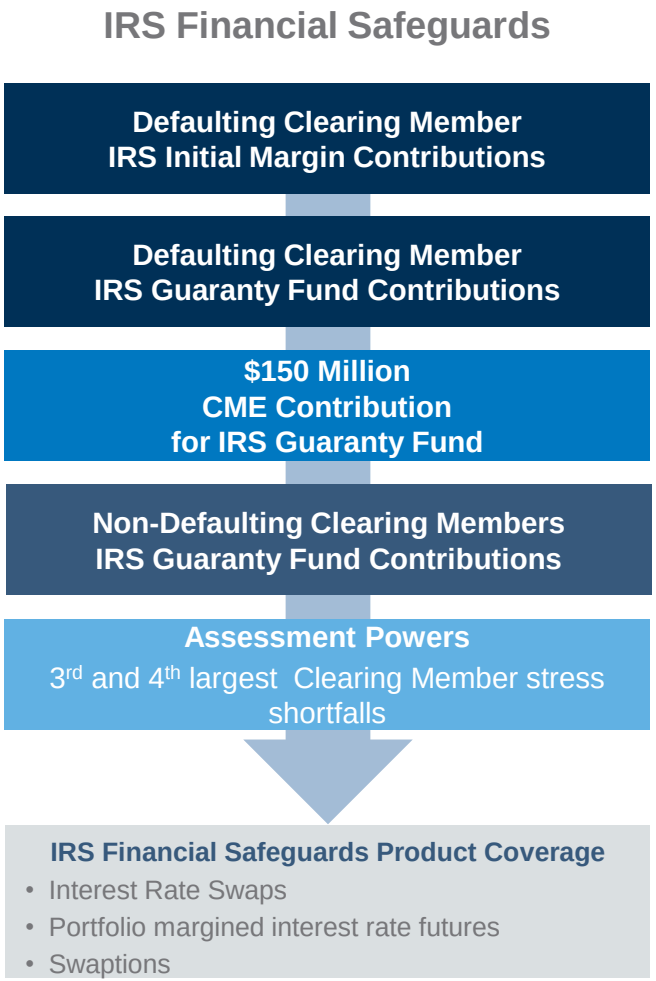
Guaranty Fund Contributions

- Each IRS Clearing Member is required to contribute a minimum amount to the IRS Guaranty Fund defined as the greater of:
  - $15 million; or
  - Clearing Member's proportional share of the fund based on:
    a) 30-day trailing average of their Portfolio Residual Loss (90%)
    b) 30-day trailing average of their Gross Notional (10%)

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IRS Financial Safeguards Waterfall

IRS Financial Safeguards



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IRS Financial Safeguards Sizing

IRS Financial Safeguards

Step 1: Calculate Net Debtor Profiles	Stress Loss – Margin Collateral = Stress Shortfall
Step 2: Identify Four Largest Net Debtors	1st and 2nd Largest Net Debtor = Guaranty Fund 3rd and 4th Largest Net Debtor = Assessments
Step 3: Calculate Each IRS Clearing Member Guaranty Fund Contribution	Each Clearing Members pro-rata share of the IRS Guaranty Fund
Step 4: Calculate Each IRS Clearing Member Potential Assessments	Each Clearing Member's potential assessment accounting for the default of any four Clearing Members from the five Clearing Members with the largest stress shortfalls

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Stress Testing Overview

IRS Financial Safeguards

- Stress testing is designed to ensure that the Guaranty Fund is sufficiently sized to cover losses during large systemic events that could result in losses that would exceed available margin collateral
- CME Clearing's stress testing is overseen by a comprehensive governance structure designed to ensure thorough review and scrutiny of stress scenarios and procedures

Methodology

Identify Historical Scenarios

- Stress testing scenarios are identified to reflect financial market crises over the past four decades
- The key indicator for these scenarios is the sudden uptick of the realized volatility of the yield curve

Construct Synthetic Scenarios

- Synthetic scenarios are utilized to supplement historical scenarios that represent macro- and micro-shocks

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Customer Resources

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CME CORE: Clearing Online Risk Engine

Customer Resources

CME CORE is an interactive margin calculator that enables users to calculate and evaluate initial margin requirements for all CME Group Products and efficiently manage risk using powerful OTC analytics

Supported CME Clearing products

- Listed derivatives
- Interest rate swaps and cross-margined interest rate futures
- Credit Default Swaps
- OTC FX
- ***NEW!*** CME Europe products

Accessing CME Core

- Sign-up for free at https://cmecore.cmegroup.com/core
- E-mail CORE team at cme.core@cmegroup.com to obtain access to OTC margin calculators

Key Benefits

- Timely margin funding information for trade execution and reconciliation
- Effectively mitigate risk
- Interaction with CME Clearing margin models
- Simple portfolio upload and fast margin results
- Download latest SPAN and CME Optimizer versions of software
- API connectivity available
- Deployable IRS and CDS software available
- ***NEW!*** Real-time margin dashboard and positions features

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